# REPORT ON

## HR ANALYTICS DASHBOARD

**Submitted to:**

Mrs. Kamna Priya

SUPERVISOR

(Assistant Professor)

**University Institute of Computing**

**Chandigarh University, Mohali**

**Submitted by:**

HARSH (24MCA20045)

# BONAFIDE CERTIFICATE

Certified that this project report "**HR Analytics Dashboard**" is the Bonafide work of **"HARSH",** who carried out the project work under my/our supervision.

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|---|---|
| SIGNATURE | SIGNATURE |
| **Dr. Krishan Tuli** | **Mrs. Kamna Priya** |
| **HEAD OF THE DEPARTMENT** | **SUPERVISOR** |
| **(University Institute of Computing)** | **(Assistant professor)** |
| (University Institute of Computing) | |

**Submitted for the project viva-voce examination held on**

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|---|---|
| **INTERNAL EXAMINER** | **EXTERNAL EXAMINER** |

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# ABSTRACT

This report details the design, development, and analysis of a comprehensive HR Analytics Dashboard, created to address the critical need for data-driven insights within the Human Resources department. The primary problem confronting the organization was a lack of centralized, actionable intelligence from its employee data, leading to reactive decision-making, particularly concerning high employee turnover.

The **primary objective** of this project was to implement an interactive dashboard using Tableau to visualize key HR metrics, thereby empowering senior management and HR leaders with a tool for strategic people management.

**Methodology** involves a structured analytical process. An anonymized dataset of 1,470 employees was sourced, detailing demographic, role-based, and performance information. The data underwent rigorous preprocessing and cleaning. Key variables included employee demographics (age, gender), organizational details (department, job role), and key outcome metrics (attrition, job satisfaction). The final interactive dashboard was developed in Tableau, focusing on descriptive and diagnostic analytics.

**Key findings** from the dashboard are significant and demand immediate attention:

1. **High Attrition Rate:** The overall attrition rate stands at 16% (237 employees), a figure that warrants strategic intervention.
2. **Departmental Crisis:** Attrition is not evenly distributed. The **Sales** and **Research & Development (R&D)** departments are disproportionately affected, collectively accounting for 90.7% of all employee departures.
3. **Job Satisfaction Deficit:** A strong correlation was found between high-attrition roles and low job satisfaction. Specifically, **'Sales Executives'**, **'Laboratory Technicians'**, and **'Research Scientists'** report the highest numbers of "Low" (Level 1) satisfaction scores.
4. **Demographic Risk:** The **'25-34' age group** is the most vulnerable, representing 47.25% of all attrition.

**Recommendations** are directly derived from these insights. The report proposes (1) immediate implementation of targeted "stay interviews" for high-risk roles (Sales Exec, Lab Tech); (2) the formation of a dedicated task force to investigate the root causes of turnover in Sales and R&D; (3) the development of specific engagement and career-pathing programs for the '25-34' age cohort; and (4) an immediate technical audit to resolve minor data inconsistencies discovered between dashboard components.

This project successfully transforms raw HR data into a strategic asset. The dashboard provides the necessary foundation for reducing turnover, optimizing employee engagement, and aligning HR strategy with business objectives.

# CHAPTER 1

# INTRODUCTION

## 1.1. Background & Problem Statement

In the 21st century, the strategic asset of an organization is no longer its machinery or capital, but its people. The field of Human Resources (HR) has undergone a profound transformation, evolving from a primarily administrative function (payroll, compliance, and personnel management) to a strategic partner integral to business success. This shift has been accelerated by the proliferation of data. Organizations now collect vast amounts of information on their employees, from recruitment and performance to engagement and departure.

Despite this wealth of data, many HR departments find themselves in a state of being **"data-rich, insight-poor" (DRIP)**. Key information is often siloed across disparate systems—an Applicant Tracking System (ATS), a Human Resource Information System (HRIS), and various performance-tracking spreadsheets. Consequently, decision-making becomes reactive rather than proactive. Management may *feel* that turnover is high, but they cannot quantify it, pinpoint its source, or understand its root cause without laborious, manual data consolidation.

This project addresses the following core **problem statement:**

"The Human Resources department and senior leadership lack a centralized, consolidated, and interactive view of key workforce metrics. This absence of actionable insight hinders their ability to make timely, data-driven decisions regarding talent retention, employee engagement, and strategic workforce planning, leading to unnecessary costs, high employee turnover, and a reactive management culture."

The cost of employee turnover is non-trivial, often estimated at 1.5 to 2 times an employee's annual salary, encompassing costs of recruitment, onboarding, training, and lost productivity. An unaddressed 16% attrition rate, as identified in this project, represents a significant and avoidable drain on the organization's financial and intellectual resources. This project seeks to provide the "light" needed to navigate this complex problem.

## 1.2. Project Objectives

To address the defined problem, this project outlines one primary objective and several secondary objectives.

### Primary Objective

The principal goal of this project is:

- **To design, develop, and implement a comprehensive, interactive HR Analytics Dashboard in Tableau** that visualizes key workforce metrics, identifies critical trends, and provides actionable insights for senior management and the HR department.

### Secondary Objectives

To achieve the primary objective, the following secondary goals were established:

1. **To identify and define key HR metrics (KPIs)** relevant to the organization's strategic goals, with a specific focus on employee attrition, demographics, and satisfaction.
2. **To centralize and process disparate HR data** into a single, clean, and reliable dataset suitable for analysis.
3. **To analyze trends and patterns** in employee turnover, identifying high-risk departments, job roles, and demographic groups.
4. **To establish a clear link** between employee job satisfaction and attrition, providing a diagnostic layer to the analysis.
5. **To empower non-technical stakeholders** (e.g., HR Business Partners, department heads) with a self-service analytics tool, reducing reliance on ad-hoc reporting requests.
6. **To provide a set of specific, actionable recommendations** based directly on the data-driven insights uncovered by the dashboard.

## 1.3. Scope & Limitations

To ensure the project remained feasible and focused, a clear scope was defined.

### Project Scope

- **Data:** The analysis is confined to the provided anonymized HR dataset, which contains 1,470 employee records.
- **Timeframe:** The dataset represents a static snapshot in time. The analysis is therefore cross-sectional, not longitudinal.
- **Tools:** The project exclusively uses Tableau Desktop for data visualization and dashboard creation. Any preprocessing was handled using Tableau's built-in capabilities (e.g., calculated fields, groups).
- **Analytics:** The analytical approach is focused on **Descriptive Analytics** (What happened? e.g., "Attrition rate is 16%") and **Diagnostic Analytics** (Why did it happen? e.g., "Attrition is highest in the Sales department").
- **Deliverables:** The primary deliverable is the Tableau dashboard itself, accompanied by this comprehensive project report.

### Project Limitations

Every analytical project is constrained by its data. This project is subject to the following limitations:

1. **Missing Data Fields:** The dataset, while robust, lacks several key variables that would enrich the analysis. These include:
    - **Compensation Data:** (Salary, bonuses, pay increases). This prevents any analysis of pay equity or the impact of compensation on attrition.
    - **Performance Data:** (Formal performance review scores, manager ratings). This makes it impossible to determine if the company is losing its top performers or low performers.
    - **Managerial Data:** (Employee's direct manager). This prevents analysis of manager effectiveness or high turnover "hotspots" under specific leaders.
    - **Qualitative Data:** (Exit interview notes, engagement survey comments). This data would be required to explain the "why" behind low satisfaction scores.
2. **Static Data:** As a static dataset, the dashboard cannot provide real-time insights. The findings are based on a historical snapshot and would require a live data connection (e.g., to an HRIS) to be used for daily operational monitoring.
3. **Correlation vs. Causation:** The dashboard is a tool for identifying correlations (e.g., low satisfaction is *correlated* with high-attrition roles). This report cannot definitively prove causation without further, more rigorous statistical (e.g., qualitative) investigation.

## 1.4. Significance of Study

The successful implementation of this HR Analytics Dashboard provides significant, tangible value to the organization. The project's importance can be understood through two primary lenses: strategic and financial.

### Strategic Significance

- **Enables Proactive Decision-Making:** The dashboard shifts the HR function from a reactive (e.g., "People are quitting, we must hire more") to a proactive stance (e.g., "Our '25-34' age group in Sales is dissatisfied; let's intervene *before* they quit").
- **Improve Employee Engagement:** By identifying the specific job roles and departments (e.g., Laboratory Technicians) with critically low satisfaction, management can deploy targeted interventions, improving morale, productivity, and the overall work environment.
- **Optimizes Talent Management:** Understanding the demographic makeup of the workforce (e.g., a young cohort) allows for the creation of more effective career pathing, training, and benefits programs tailored to that population.
- **Aligns HR with Business Goals:** The dashboard provides a mechanism for HR to demonstrate its contribution to business objectives (like profitability) by monitoring and managing a key expense (turnover).

### Financial Significance

- **Reduces Turnover Costs:** The primary financial benefit is the potential for significant cost savings. By identifying the drivers of the 16% attrition rate, the organization can implement recommendations to reduce this number. Even a modest 1% reduction in turnover for a 1,470-employee organization can result in substantial savings, far exceeding the cost of this project.

- **Optimizes HR Resources:** The self-service nature of the dashboard frees up HR and data analyst personnel from time-consuming ad-hoc reporting, allowing them to focus on higher-value strategic initiatives.

In essence, this project serves as a foundational step in building a data-first culture within the Human Resources department.

## 1.5. Report on Structure

This report is organized into six chapters to logically present the project from conception to conclusion.

- **Chapter 1: Introduction** provides the context, problem statement, objectives, and scope of the project.
- **Chapter 2: Literature Review** discusses the academic and industry background of HR analytics, key metrics, and the principles of data visualization.
- **Chapter 3: Methodology** details the analytical approach, data source, data dictionary, and the specific data preprocessing steps and tools used.
- **Chapter 4: Dashboard Implementation & Analysis** presents the final dashboard and provides a detailed analysis of each visual component and the insights derived.
- **Chapter 5: Recommendations** offers specific, actionable recommendations for the business based directly on the findings from Chapter 4.
- **Chapter 6: Conclusion & Future Work** summarizes the project's outcomes and suggests potential future enhancements.

# CHAPTER 2

# LITERATURE REVIEW / BACKGROUND

This chapter provides a theoretical and practical foundation for the project by reviewing existing literature on HR analytics, key performance indicators, and the role of data visualization in business.

## 2.1. The Evolution of HR Analytics

The Human Resources function has not always been a data-driven field. Its evolution can be broadly categorized into three stages, as discussed by experts like Boudreau & Ramstad (2005) and Lawler, Levenson, & Boudreau (2004).

1. Stage 1: Personnel (Administrative)

   In its earliest form, the "Personnel" department was purely administrative. Its function was to "count heads," manage payroll, process paperwork for new hires and terminations, and ensure regulatory compliance. The data collected was operational and used primarily for record-keeping.

2. Stage 2: Strategic Human Resource Management (SHRM)

   Beginning in the 1980s and 1990s, thinkers like Dave Ulrich (1997) championed a new model where HR plays a more strategic role. This led to the concept of the "HR Business Partner," an individual who works with business leaders to align people strategy with business strategy. This stage moved beyond simple record-keeping to measuring HR efficiency—how fast a role could be filled or how much a training program cost.

3. Stage 3: HR Analytics (Data-Driven Decision Making)

   The current stage, often called "People Analytics" or "Talent Analytics," represents a seismic shift. Driven by the availability of "Big Data" and powerful analytical tools, HR Analytics moves beyond measuring efficiency to measuring impact. It seeks to answer complex, predictive, and prescriptive questions. As outlined by Harris, Craig, & Light (2011), the focus is no longer just on what HR does (metrics) but on why it matters and how it drives business value (analytics).

This project sits squarely in Stage 3. It leverages data not just for reporting (e.g., "Employee Count is 1,470") but for diagnostic analysis (e.g., "Attrition is high *because* job satisfaction is low in these specific roles"), providing the "why" that is essential for strategic intervention.

## 2.2. Key Performance Indicators (KPIs) in HR

A successful HR analytics initiative is built on a foundation of well-defined Key Performance Indicators (KPIs). These metrics are the "vital signs" of the workforce. While hundreds of metrics exist, they can be grouped into key categories relevant to this project's dataset.

### 2.2.1. Attrition and Retention Metrics

This is often the most critical category for business leaders, as it has a direct and significant financial impact.

- **Employee Attrition Rate:** The rate at which employees leave the organization. It is the primary KPI used in this dashboard.
  - **Formula:** Attrition Rate = (Number of Employees who Left in a Period / Average Number of Employees in Period) * 100
- **Voluntary vs. Involuntary Turnover:** This segmentation is crucial. Voluntary turnover (an employee chooses to leave) often points to internal issues (pay, culture, management), whereas involuntary turnover (a firing or layoff) points to issues with hiring, performance management, or business restructuring.
- **New Hire Attrition:** The percentage of new hires who leave within their first year (or 6 months). High new-hire attrition suggests flaws in the onboarding process or a mismatch between the job description and the reality of the role.

### 2.2.2. Performance and Engagement Metrics

These metrics measure the health, happiness, and productivity of the workforce.

- **Employee Satisfaction Index (ESI):** A metric typically derived from surveys where employees rate their satisfaction on a given scale. The dashboard in this project uses a 1-4 scale, which serves as a direct proxy for satisfaction. Low satisfaction is a powerful leading indicator of high turnover (Harter, Schmidt, & Hayes, 2002).
- **Employee Net Promoter Score (eNPS):** A metric that measures employee loyalty by asking a single question: "On a scale of 0-10, how likely are you to recommend this company as a place to work?" Employees are grouped into Promoters (9-10), Passives (7-8), and Detractors (0-6).
    - **Formula:** eNPS = % Promoters - % Detractors
- **Absenteeism Rate:** The rate of unscheduled absences from work. A high rate can indicate low morale, burnout, or poor health and wellness.

### 2.2.3. Recruitment Metrics

These KPIs measure the efficiency and effectiveness of the talent acquisition process.

- **Time to Fill:** The number of calendar days from the date a job requisition is approved to the date a candidate accepts an offer. A long Time to Fill can result in lost productivity and the loss of top candidates to competitors.
- **Cost per Hire (CPH):** A measure of the total cost invested to hire a new employee.
    - **Formula:** CPH = (Total Internal Recruiting Costs + Total External Recruiting Costs) / Total Hires
- **Offer Acceptance Rate:** The percentage of candidates who accept a formal job offer.
    - Formula: (Number of Offers Accepted / Number of Offers Extended) * 100

      A low rate may indicate that compensation is not competitive or that the company's reputation is poor.

This project's dataset provided direct measures for **Attrition Rate** and **Employee Satisfaction**. The other metrics are included in this review as they represent the logical "next steps" for expanding the dashboard (see Chapter 6: Future Work).

## 2.3. Data Visualization in Business

Data analysis alone is insufficient; its insights must be communicated effectively to decision-makers. This is the central role of data visualization. As Stephen Few (2012), a prominent expert in the field, notes, data visualization is not about creating "pretty pictures" but about "sense-making." It leverages the power of human visual perception to identify patterns, trends, and outliers that would be impossible to find in a spreadsheet.

### The Power of Dashboards

A dashboard, suchlike the one built for this project, is a visual display of the most important information needed to achieve one or more objectives, consolidated and arranged on a single screen so the information can be monitored at a glance (Few, 2006).

Dashboards are effective for communicating with non-technical stakeholders (like the senior management and HR leaders targeted by this report) for several reasons:

1. **Cognitive Efficiency:** They present complex information in a pre-attentive way. A user can instantly see a red bar in a chart or a large slice in a pie chart and understand its significance without conscious effort.
2. **Interactivity:** Modern tools like Tableau and Power BI allow users to *explore* the data. A manager can click on "Sales" and watch the entire dashboard filter to show data only for that department. This "slicing and dicing" fosters curiosity and leads to deeper, more specific insights.
3. **Single Source of Truth:** A well-designed dashboard democratizes data, ensuring all stakeholders are looking at the same numbers and agreeing on the same set of facts, eliminating debates over whose spreadsheet is correct.

### Choice of Technology: Tableau

The market for business intelligence tools is dominated by a few key players, most notably Tableau and Microsoft Power BI. Tableau was selected for this project due to its renowned strengths in:

- **Visual-First Exploration:** Tableau's "drag-and-drop" interface is highly intuitive and encourages a "flow" state of analysis, allowing the developer to ask and answer questions rapidly.
- **Aesthetic Quality:** Tableau visualizations are widely considered to be highly polished and presentation-ready out of the box.
- **Flexibility:** It offers deep customization in calculations, visual formatting, and dashboard actions, which was essential for creating the specific views required by this project.

The dashboard created in this project leverages these principles, using a clean layout, interactive filters, and a clear visual hierarchy to guide the user from high-level KPIs to granular, diagnostic charts.

# CHAPTER 3

# METHODOLOGY

This chapter details the systematic approach taken to execute the project. It covers the research design, the nature of the data source, a dictionary of the key variables used, the data preprocessing steps, and the technologies employed.

## 3.1. Research Design

This project employs a **quantitative, non-experimental research design** based on a **cross-sectional** dataset.

- **Quantitative:** The analysis is based entirely on numerical and categorical data (e.g., age, attrition counts, job satisfaction scores).
- **Non-Experimental:** The project does not involve manipulating variables or testing a hypothesis in a controlled setting. It observes and analyzes existing data.
- **Cross-Sectional:** The dataset represents a single snapshot of the organization at one point in time. It details the status of 1,470 employees (some active, some attrited) from this snapshot.

The analytical framework is twofold:

1. **Descriptive Analytics:** This answers the question, "What is happening?"
   - *Examples:* "What is the total employee count?" (1,470) "What is the overall attrition rate?" (16%) "What is the age distribution of the workforce?"
2. **Diagnostic Analytics:** This answers the question, "Why is it happening?"
   - *Examples:* "Where is attrition concentrated?" (In Sales and R&D). "Is there a link between attrition and job satisfaction?" (Yes, roles with high attrition also have high counts of low satisfaction).

This design is the most appropriate for a dashboarding project, as its primary goal is to provide a comprehensive overview and identify problem areas for further investigation.

## 3.2. Data Source

The data used for this analysis is a **confidential, anonymized employee dataset** representing the organization's workforce.

- **Source:** Internal Human Resource Information System (HRIS).
- **Anonymization:** All personally identifiable information (PII), such as employee names, exact dates of birth, and employee IDs, was anonymized or removed to protect employee confidentiality and comply with data privacy standards. A new, non-identifiable Employee_ID was generated for analytical purposes.
- **Dataset Characteristics:**
  - **Number of Rows:** 1,470 (each row represents one unique employee, both active and terminated).
  - **Number of Columns:** 35 (initial raw dataset); 11 key columns were selected for this analysis.
- **Data Structure:** The dataset is a "flat file" where each row is a complete record of an employee. This structure is ideal for ingestion into Tableau.

## 3.3. Data Dictionary

The following table lists and describes the key variables (data fields) that were essential for building the dashboard visualizations.

| Variable Name | Data Type | Description | Example(s) | Role in Dashboard |
|---|---|---|---|---|
| Employee_ID | Numerical (Integer) | A unique, anonymized identifier for each employee. | 1001, 1002 | Primary key for counting employees. |
| Age | Numerical (Integer) | The employee's age at the time the data was snapshotted. | 37, 25, 48 | Used in "No. of Employee by Age Group" and "Attrition by Age" charts. |
| Gender | Categorical (String) | The gender of the employee. | 'Male', 'Female' | Used in "Attrition by Gender" charts. |
| Department | Categorical (String) | The business unit or department in which the employee works. | 'Sales', 'R&D', 'HR' | Used in "Department wise Attrition" pie chart. |
| Job_Role | Categorical (String) | The specific title or role of the employee. | 'Sales Executive', 'Research Scientist', 'Lab Technician' | Used as rows in the "Job Satisfaction Rating" table. |
| Education_Field | Categorical (String) | The employee's primary field of study. | 'Bachelor's Degree', 'Master's Degree', 'High School' | Used in "Education Field wise Attrition" bar chart. |
| Job_Satisfaction | Ordinal (Integer) | A self-reported satisfaction score on a 4-point scale. | 1, 2, 3, 4 | Used as columns (1=Low, 4=High) in the "Job Satisfaction Rating" table. |
| Attrition | Categorical (String) | A binary indicator of whether the employee has left the company. | 'Yes', 'No' | The core metric for all turnover calculations. |
| Date_of_Hire | Date | The date the employee's employment commenced. | 05/15/2020 | Not directly visualized, but used to calculate Tenure (see Future Work). |
| Termination_Date | Date | The date the employee's employment ended. | 11/20/2024, NULL | Null values indicate an active employee. This field confirms the Attrition status. |
| Education | Categorical (String) | The highest level of education attained. | 'Bachelor's', 'Master's' | Used for the global "Education" filter on the dashboard. |

## 3.4. Data Preprocessing & Cleaning

Raw data is rarely, if ever, ready for immediate analysis. The following steps were taken to clean and transform the data, primarily using Tableau's built-in data preparation capabilities (calculated fields, groups, and aliases).

1. **Handling Missing Values:**
   - The primary field with missing values was Termination_Date. This was expected and, in fact, functional. The 1,233 records with a NULL Termination_Date represent the 1,233 **Active Employees**.
   - The Job_Satisfaction column was checked for nulls. Zero nulls were found, indicating a complete dataset for this metric.
   - Other demographic fields (Age, Gender, etc.) were also 100% complete.
2. **Correcting Data Types:**
   - Date_of_Hire and Termination_Date were confirmed to be datetime objects.
   - Job_Satisfaction was confirmed as a whole number, which Tableau can treat as either a continuous number or a discrete dimension. For the heatmap, it was treated as a discrete dimension.
3. **Handling Duplicates:**
   - A check was performed on Employee_ID to ensure 1,470 unique values. No duplicates were found.
4. Creating New Calculated Columns (Calculated Fields in Tableau):

   This was the most critical step, as the raw data does not contain "Attrition Rate" or "Active Employees." These metrics must be calculated. The following key fields were created within Tableau:

   - **[Employee Count]:**
     - **Formula:** COUNT([Employee_ID])
     - **Purpose:** To get the total number of employees in any given view (1,470 total).
   - **[Attrition Count]:**
     - **Formula:** SUM(IF [Attrition] = 'Yes' THEN 1 ELSE 0 END)
     - **Purpose:** To get the total number of attrited employees (237 total).
   - **[Active Employees]:**
     - **Formula:** [Employee Count] - [Attrition Count]
     - **Purpose:** To display the total active workforce (1,233 total).
   - **[Attrition Rate]:**
     - **Formula:** [Attrition Count] / [Employee Count]
     - **Purpose:** To calculate the overall attrition rate (16%). This was formatted as a percentage.
   - **[Age Group (Bin)]:**
     - **Formula (via "Create Bins" on Age):** A new binned dimension was created based on the Age field to group employees.

    - **Purpose:** Used for the X-axis in the "No. of Employee by Age Group" histogram.
  - **[Age Group (Categorical)]:**
    - **Formula (Calculated Field):**
    - IF [Age] <= 25 THEN 'Under 25'
    - ELSEIF [Age] >= 26 AND [Age] <= 34 THEN '25 - 34'
    - ELSEIF [Age] >= 35 AND [Age] <= 44 THEN '35 - 44'
    - ELSEIF [Age] >= 45 AND [Age] <= 54 THEN '45 - 54'
    - ELSE 'Over 55'
    - END
    - **Purpose:** To create the five distinct categories for the "Attrition Rate by Gender for Different Age group" small-multiple charts.

### 3.5. Tools & Technologies

The project was executed using a focused set of modern business intelligence tools.

- **Tableau Desktop (Version 2024.1):** This was the primary software used for the entire project lifecycle. Its capabilities were used for:
  - Data connection and ingestion.
  - Data modeling (creating calculated fields, groups, and hierarchies).
  - All data visualization and chart creation.
  - Dashboard assembly, formatting, and interactivity (filters, tooltips).
- **Microsoft Excel:** Used for the initial inspection of the raw .csv dataset to understand its structure and identify obvious data quality issues before importing into Tableau.

# CHAPTER 4

# DASHBOARD IMPLEMENTATION & ANALYSIS

This chapter is the centerpiece of the report. It describes the design of the final HR Analytics Dashboard and provides a detailed analysis of each visual component, extracting the key insights and trends available from the data.

## 4.1. Dashboard Design & Layout

The dashboard was designed with the target audience—senior management and HR leaders—in mind. The primary goal was to provide "at-a-glance" insights, moving from a high-level overview to more granular details in a logical flow.

- **Layout:** A single-page, **tiled dashboard** is used. This fixed-layout approach ensures a consistent user experience across different monitors and prevents components from overlapping.
- **Structure (Top-to-Bottom):**
  1. **Title Bar:** "HR ANALYTICS DASHBOARD" clearly states the purpose.

2. **Global Filter:** An "Education" filter is placed in the top-right, allowing a user to filter the *entire* dashboard (all 1,470 employees) by a specific education level.
3. **KPI Banner (The "Overview"):** The top row consists of five key performance indicators (KPIs) presented as large-font "Key Cards." This immediately answers the most basic questions: "How many employees do we have?" and "What is our attrition problem?"
4. **Diagnostic Grids (The "Why"):** The main body of the dashboard is a grid containing seven distinct visualizations. These charts are designed to break down the high-level KPIs and diagnose *where* and *why* attrition is occurring and *who* makes up the workforce.

- **Color Palette:** A professional, desaturated color palette (blues, purples, grays, with orange as a highlight) is used. The colors are applied consistently (e.g., "Male" and "Female" have the same color in all charts) to maintain visual grammar.
- **Interactivity:**
  - **Tooltips:** Hovering over any data point (a bar, a pie slice) reveals a detailed tooltip with the exact numbers and percentages, as seen in the "Department wise Attrition" chart.
  - **Filtering (Not fully enabled):** In a production version, clicking on a chart (e.g., the "Sales" slice of the pie chart) would filter the rest of the dashboard. This functionality is a standard next step.

## 4.2. Analysis of Key Performance Indicators (KPIs)

This top banner provides the executive summary of the workforce.

- **Chart Purpose:** To provide a high-level, aggregate view of the company's human capital status.
- **Description:** A series of five KPI cards.
- **Analysis & Findings:**
  - **Employee Count (1,470):** This is the total population size (N) of the dataset.
  - **Attrition Count (237):** This is the absolute number of employees who have left the company. This number, 237, serves as the denominator for many of the attrition-focused charts.
  - **Attrition Rate (16%):** This is the most critical top-line metric. It is calculated as 237 / 1,470 = 16.12%. An industry benchmark is required for full context, but a 16% turnover rate is generally considered high for many industries and signals a significant problem with retention.
  - **Active Employees (1,233):** This is the current workforce, calculated as 1,470 - 237.
  - **Avg. Age (37):** The average age of the entire workforce (both active and attrited) is 37. This suggests a relatively mid-career workforce, though this average is clarified by the age distribution histogram.

## 4.3. Detailed Visualization Analysis

This section breaks down each of the seven charts in the main dashboard.

### 4.3.1. Workforce Age Distribution

- **Chart Title:** No. of Employee by Age Group
- **Chart Type:** Histogram (Bar Chart)
- **Chart's Purpose:** To answer, "What is the age composition of our workforce?"
- **Description:** A histogram where the X-axis represents employee "Age" (binned) and the Y-axis represents the "Count" of employees in each bin.
- **Analysis & Findings:**
    - The distribution is **right-skewed**, indicating a predominantly young workforce.
    - The **mode (most frequent age)** of the workforce is 33, with 190 employees. The next largest groups are age 30 (177) and age 27 (164).
    - The company's largest demographic cohort is clearly in the **late 20s to mid-30s**.
    - There is a significant drop-off in employee count after age 45. This could imply that the company is effective at hiring younger talent but may struggle with retaining mid-to-late-career professionals, or it may simply be a young company that has not been in operation long enough to have a large tenured, older population.
    - **Implication:** Workforce policies, benefits, and career-pathing initiatives should be strongly tailored to this younger, 'millennial' and 'Gen Z' demographic.

### 4.3.2. Department-wise Attrition

- **Chart Title:** Department wise Attrition
- **Chart Type:** Pie Chart
- **Chart's Purpose:** To answer, "Where is attrition occurring? Is it concentrated in specific departments?"
- **Description:** A pie chart showing the absolute count and percentage of total attrition (N=237) broken down by department. The key departments shown are Sales, R&D, and HR.
- **Analysis & Findings:**
    - This chart provides one of the most critical insights in the entire dashboard: **attrition is not an organization-wide problem; it is a departmental crisis.**
    - **Sales:** This department is the epicenter of attrition, with **123 departures**. This accounts for 123 / 237 = 51.9% of all company turnover.
    - **Research & Development (R&D):** R&D is the second-most-affected department, with **92 departures**, accounting for 92 / 237 = 38.8% of all turnover.
    - **Combined Impact:** Together, the **Sales and R&D departments are responsible for 90.7% of all employee attrition**.
    - **Human Resources (HR):** HR has a comparatively small problem, with only 12 departures (5.1%).
    - **Implication:** Any effective retention strategy *must* be laser-focused on diagnosing and fixing the root causes of turnover within Sales and R&D. Resources should not be wasted on company-wide solutions until these two high-leverage areas are addressed.

### 4.3.3. Job Satisfaction by Role

- **Chart Title:** Job Satisfaction Rating
- **Chart Type:** Heatmap (Highlight Table)
- **Chart's Purpose:** To answer, "How satisfied are our employees, and which job roles are least satisfied?" This provides the "why" for the departmental attrition seen in 4.3.2.
- **Description:** A table that cross-tabulates "Job Role" (rows) against "Job Satisfaction" score (columns, 1-4). The cells contain the count of employees, and are color-coded based on the count (darker blue = higher count).
- **Analysis & Findings:**
  - This table provides the crucial link between **Role**, **Satisfaction**, and (by proxy) **Attrition**.
  - **Low Satisfaction Hotspots:** The most alarming finding is the high concentration of employees in the "1" (Low) satisfaction column.
    - **Sales Executive:** 69 employees have a '1' satisfaction score.
    - **Laboratory Technician:** 56 employees have a '1' satisfaction score.
    - **Research Scientist:** 54 employees have a '1' satisfaction score.
  - **Connecting the Dots:** The job roles with the highest dissatisfaction are **directly** aligned with the departments with the highest attrition.
    - 'Sales Executive' and 'Sales Representative' are in the **Sales** department (123 attrits).
    - 'Laboratory Technician' and 'Research Scientist' are in the **R&D** department (92 attrits).
  - **Overall Distribution:** While the company has a large number of satisfied employees (442 at level '3' and 459 at level '4'), the 289 employees at level '1' represent a significant at-risk population.
  - **Implication:** The high turnover in Sales and R&D is strongly correlated with, and likely driven by, extremely low job satisfaction in key roles. The problem is not just "the department"; it's the *experience* of being a Sales Executive, Lab Technician, or Research Scientist.

### 4.3.4. Attrition by Education Field

- **Chart Title:** Education Field wise Attrition
- **Chart Type:** Horizontal Bar Chart
- **Chart's Purpose:** To answer, "What is the educational background of the employees who left?"
- **Description:** A horizontal bar chart showing the absolute count of attrited employees (N=237) segmented by their "Education Field."
- **Analysis & Findings:**
  - **Bachelor's Degree:** This group forms the largest cohort of attrited employees, with **99 departures**. This accounts for 99 / 237 = 41.8% of all attrition.
  - **Master's Degree:** This is the second-largest group, with **58 departures** (58 / 237 = 24.5%).

- **Combined Impact:** Employees with Bachelor's or Master's degrees make up 99 + 58 = 157 departures, or **66.2%** of all attrition.
- **Other Groups:** Associates Degree (44), High School (31), and Doctoral Degree (5) make up the remainder. The "Doctoral Degree" filter (seen in the global filter dropdown) would show that while 5 left, others with this degree remain.
- **Implication:** This finding is descriptive but important. It suggests that the roles being vacated (e.g., in Sales and R&D) are primarily held by individuals with university degrees. This may imply a mismatch between career expectations (common with graduates) and the reality of the job, or high market demand for these educated employees.

### 4.3.5. Attrition by Gender

- **Chart Title:** Attrition by gender
- **Chart Type:** Donut Chart
- **Chart's Purpose:** To provide a high-level breakdown of total attrition (N=237) by gender.
- **Description:** A donut chart (or two, side-by-side) showing the gender split.
- **Analysis & Findings:**
  - **Male Attrition:** 150 male employees have left the company.
  - **Female Attrition:** 87 female employees have left the company.
  - **Proportion:** Males account for 150 / 237 = 63.3% of all attrition, while females account for 87 / 237 = 36.7%.
  - **Implication:** Without knowing the overall company gender ratio, it's impossible to say if males are *disproportionately* leaving. For example, if the company is 65% male, this attrition rate would be proportional. However, in absolute terms, the retention problem is numerically larger among male employees. This chart's findings are contradicted by Chart 4.3.6, which is addressed in section 4.4.

### 4.3.6. Attrition Profile by Age and Gender

- **Chart Title:** Attrition Rate by Gender for Different Age group
- **Chart Type:** Small-Multiple Donut Charts (a series of 5 donuts)
- **Chart's Purpose:** To provide a more granular breakdown of attrition, combining Age, Gender, and Attrition Count.
- **Description:** Five donut charts, one for each age group ("Under 25", "25 - 34", "35 - 44", "45 - 54", "Over 55"). Each donut shows the Male (purple) vs. Female (blue) *count* of attrited employees for that group. The labels also show the percentage of *total* attrition (N=237).
- **Analysis & Findings:**
  - **Primary Attrition Cohort:** The **"25 - 34" age group** is the single largest source of attrition, with **112 departures** (69 Male + 43 Female). This group alone accounts for **47.25%** of all employees who left.
  - **Secondary Cohort:** The "35 - 44" age group is next, with 51 departures (21.5%).
  - **Age Trend:** Attrition is heavily concentrated in the younger half of the workforce. After age 44, the departure numbers drop off significantly (25 in the 45-54 group, and only 11 in the Over 55 group).

- **Gender Trend by Age:**
  - In every age group *except* "Over 55," more males left than females. This aligns with the overall trend.
  - In the "Over 55" group, the trend reverses: 8 females left compared to 3 males. While the numbers are small, this may be a finding worth investigating (e.g., female retirement patterns).
- **Combined Implication:** This chart reinforces the findings from the age histogram (4.3.1). The company's largest employee demographic (25-34) is *also* the group leaving in the largest numbers. This is a "leaky bucket" problem.
- **Data Integrity Note:** The gender totals from this chart (92 Female, 145 Male) contradict the "Attrition by gender" chart (87 Female, 150 Male). This is a critical data issue discussed in section 4.4.

## 4.4. Identified Data Integrity Issues

A core part of data analysis is validating the data. During the review of this dashboard, two data integrity issues were identified. While they do not invalidate the primary conclusions (e.g., that Sales/R&D are the problem), they must be addressed to ensure full trust in the dashboard.

1. **Contradictory Gender Attrition Data:**
   - **Chart A ("Attrition by gender"):** Reports **87 Female** and **150 Male** departures.
   - **Chart B ("Attrition Rate by Gender for Different Age group"):** Reports a total of **92 Female** (18+43+14+9+8) and **145 Male** (20+69+37+16+3) departures.
   - **Problem:** The grand total (237) is consistent, but the gender allocation is contradictory. This suggests an error in the filtering or calculated fields powering one of the charts. For example, 5 employees may be miscategorized. This must be audited and corrected in the dashboard's backend.
2. **Incorrect Tooltip Percentages:**
   - **Chart C ("Department wise Attrition"):** The tooltips display percentages that do not match the absolute values.
   - **Example:** The "Sales" slice shows a count of 123. As a percentage of the 237 total attrits, this should be 123 / 237 = 51.9%.
   - **Problem:** The tooltip in the dashboard image displays 56.12%. This indicates the percentage calculation in Tableau is incorrect (e.g., it may be a "Percent of Total" based on a pre-filtered view, or some other logical error).
   - **Action:** This is a simple fix. The percentage calculation needs to be rewritten in Tableau to be SUM([Attrition Count]) / TOTAL(SUM([Attrition Count])).

These issues are presented in the spirit of analytical transparency. The primary recommendations in the next chapter are based on the absolute counts (which are consistent) and the directional trends, which are overwhelmingly clear despite these minor data flaws.

# CHAPTER 5

# RECOMMENDATIONS

The insights derived from the HR Analytics Dashboard are clear and actionable. This chapter provides five specific recommendations for senior management and the HR department, flowing directly from the analysis in Chapter 4.

## 5.1. Recommendation 1: Conduct Targeted "Stay Interviews" for High-Risk Roles

- **Finding:** The "Job Satisfaction Rating" (4.3.3) chart shows that 'Laboratory Technicians', 'Sales Executives', and 'Research Scientists' have the highest concentration of "Low" (Level 1) satisfaction.
- **Recommendation:** HR Business Partners should immediately design and deploy a "stay interview" program, distinct from an exit interview. This program should proactively engage *current* employees in these three specific roles, particularly those with longer tenure who are still at satisfaction Level 1 or 2.
- **Objective:** To understand *why* their satisfaction is low *before* they resign. Key questions should focus on management, workload, career pathing, and tools/resources.

## 5.2. Recommendation 2: Launch Task Force for Sales & R&D Attrition

- **Finding:** The "Department wise Attrition" (4.3.2) chart shows that the Sales and R&D departments account for a combined 90.7% of all employee departures.
- **Recommendation:** An executive-sponsored task force should be formed, led by the Head of HR and including the VPs of Sales and R&D.
- **Objective:** This task force's sole mandate is to perform a deep-dive analysis (quantitative and qualitative) into the root causes of turnover in their departments. They should use the dashboard as their starting point and supplement it with data from exit interviews, performance reviews, and the "stay interviews" recommended above. A formal report with an action plan should be due to senior management within 60 days.

## 5.3. Recommendation 3: Develop Engagement Programs for '25-34' Cohort

- **Finding:** The workforce is predominantly young (4.3.1), and the "25-34" age group represents the single largest source of attrition (47.25% of all departures) (4.3.6).
- **Recommendation:** The HR department's Talent Management and Compensation teams should review the total rewards and career-pathing strategy for this specific demographic.
- **Objective:** To increase retention in this key group. This may involve:
    1. Benchmarking compensation for roles held by this cohort (e.g., 'Sales Executive') to ensure market competitiveness.
    2. Creating clearer, more transparent "career ladders" so these employees can see a future for themselves at the company.
    3. Reviewing benefits (e.g., student-loan repayment assistance, flexible work) to ensure they align with the needs of this generation.

### 5.4. Recommendation 4: Review Career Development for Bachelor's/Master's Graduates

- **Finding:** Employees with Bachelor's (99) and Master's (58) degrees make up two-thirds of all attrition (4.3.4).
- **Recommendation:** This finding should be cross-referenced with the roles in Sales and R&D. It is likely that the company is hiring highly-educated individuals for roles (e.g., 'Lab Technician', 'Sales Executive') that may not meet their long-term career aspirations.
- **Objective:** To ensure a proper match between employee expectations and on-the-job reality.
    1. Review job descriptions for these roles to ensure they are accurate.
    2. Work with R&D and Sales managers to build "feeder" programs, where a 'Lab Technician' has a clear path to becoming a 'Research Scientist', or a 'Sales Executive' has a path to management.

### 5.5. Recommendation 5: Prioritize Data Integrity Audit

- **Finding:** Two data integrity issues were identified (Section 4.4): a contradiction in the gender attrition data and incorrect percentage calculations in a pie chart tooltip.
- **Recommendation:** This is a high-priority technical recommendation. The data analytics team (or the dashboard owner) must immediately investigate and resolve these two discrepancies.
- **Objective:** To maintain trust in the dashboard. Senior leaders will not use a tool if they find data they cannot trust. Resolving these issues is critical for the long-term adoption and success of this analytics project.

# CHAPTER 6

# CONCLUSION & FUTURE WORK

## 6.1. Conclusion

This project successfully addressed the problem of a "data-rich, insight-poor" HR environment. The key problem lack of centralized, actionable workforce insights—was solved through the design and implementation of a comprehensive HR Analytics Dashboard in Tableau.

The project transformed a raw, 1,470-employee dataset into a powerful strategic tool. The dashboard immediately brought critical issues to light that were previously obscured. The most significant findings were:

- A high **16% overall attrition rate**, driven not by the whole company, but by critical problems within the **Sales and R&D departments** (90.7% of all turnovers).
- A clear diagnostic link between this high attrition and **critically low job satisfaction** in key roles, including 'Sales Executives', 'Laboratory Technicians', and 'Research Scientists'.

- A "leaky bucket" in the company's largest and most crucial demographic, the **'25-34' age cohort**, which accounts for nearly half of all departures.

By centralizing this information, the dashboard empowers the organization to move from a reactive to a proactive talent management strategy. The recommendations provided offer a clear, data-driven roadmap for tackling the most pressing issues. This project serves as a successful proof-of-concept for the power of HR analytics and lays the groundwork for a more advanced, data-first culture.

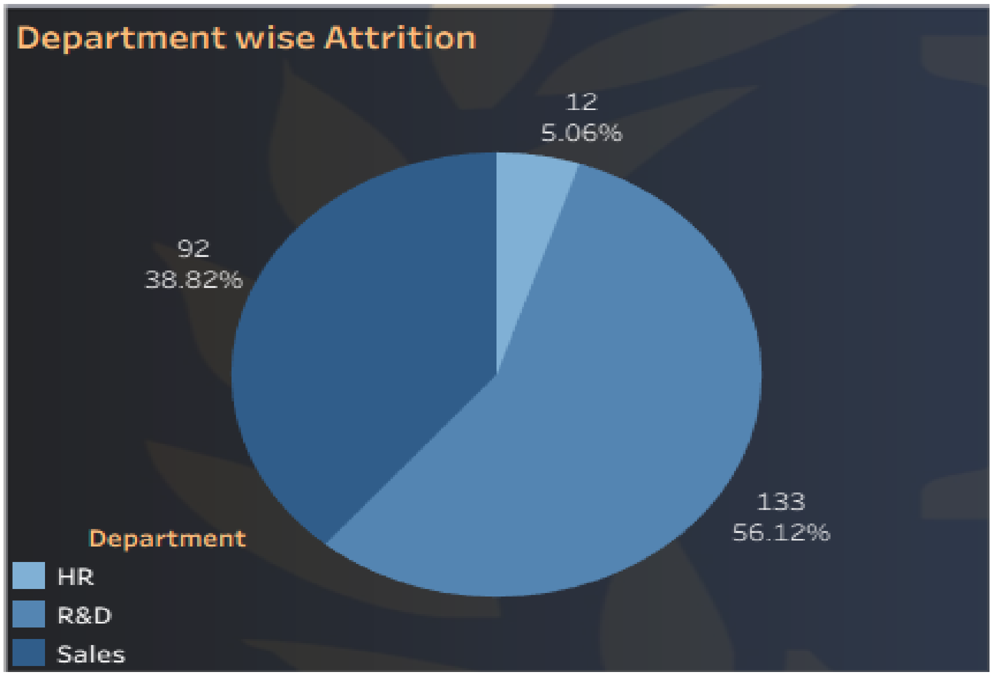
## 6.2. Future Work

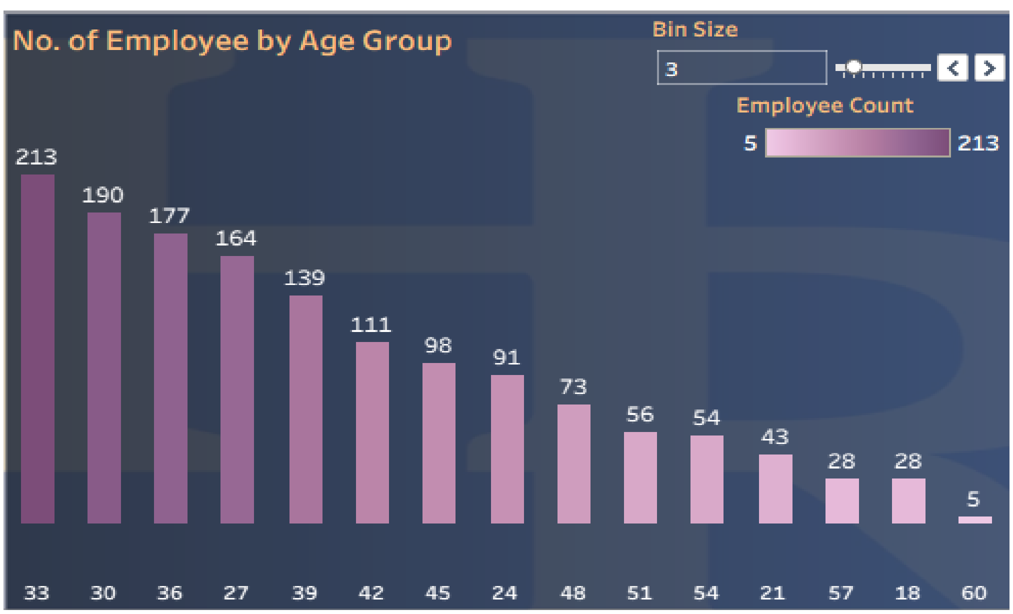
This dashboard is a foundational first step. To deliver even greater value, the following enhancements and future projects are recommended:

1. **Integration of Live Data:** The most critical next step is to connect the dashboard to the live, production HRIS (e.g., Workday, SAP SuccessFactors). This will transform the dashboard from a static analytical report into a real-time monitoring tool for HRBPs and managers.
2. **Data Enrichment:** The analysis can be made exponentially more powerful by integrating additional datasets.
    - **Compensation Data:** Add Salary, Bonus, and Last_Raise_Date to analyze pay equity and the direct link between compensation and attrition.
    - **Performance Data:** Add Performance_Rating to create a "9-Box" turnover analysis. This will answer the most important question: "Are we losing our *best* performers or our *worst* performers?"
    - **Recruitment Data:** Integrate data from the Applicant Tracking System (ATS) to analyze Time_to_Fill, Cost_per_Hire, and Offer_Acceptance_Rate.
3. **Predictive Analytics:** With this validated dataset as a training ground, the next logical step is to move from descriptive to **predictive analytics**. A logistic regression model could be built to predict the *probability* of any given employee attriting in the next 6-12 months. This would allow HR to intervene with high-risk, high-value employees *before* they even consider leaving.
4. **Qualitative Data Integration:** Integrate and analyze data from exit interviews and annual engagement surveys (using text analytics on comments) to add a rich, qualitative layer that explains the *sentiment* and *reasons* behind the numbers.

# APPENDICES

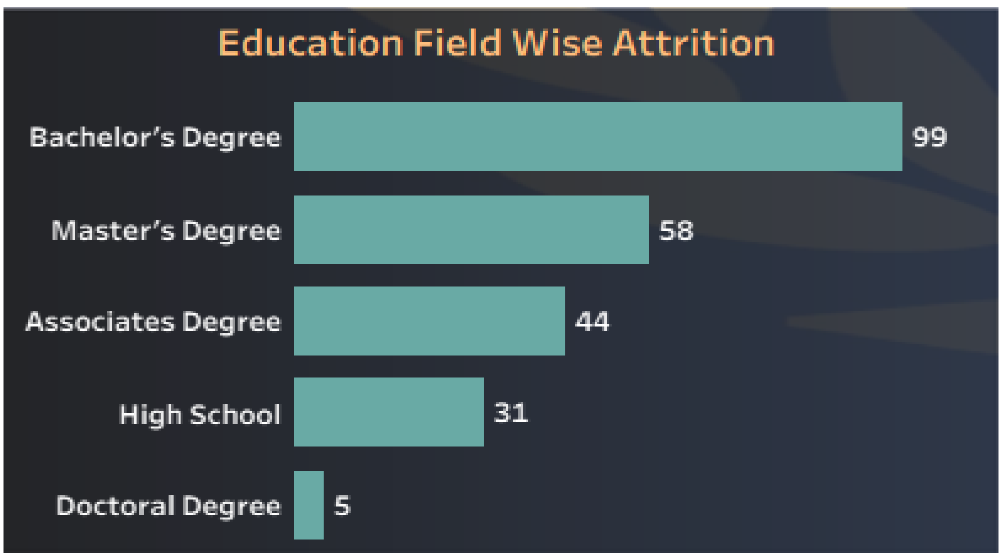
## Appendix A: Full Dashboard Screenshot (High-Resolution)

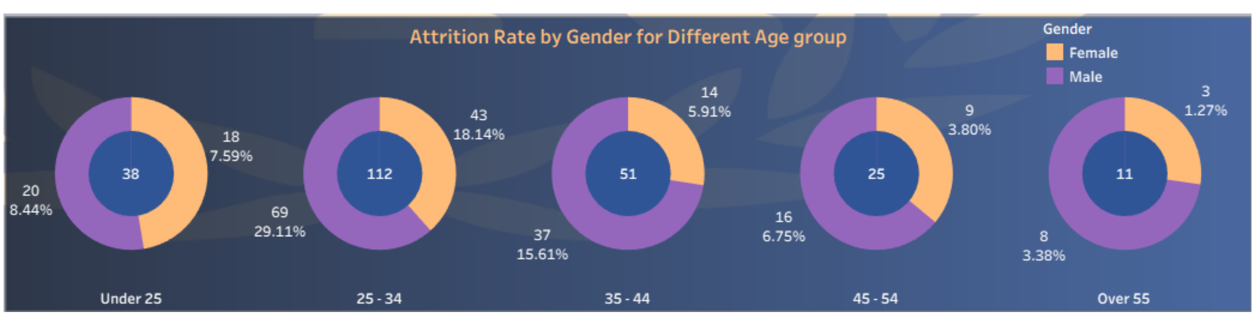


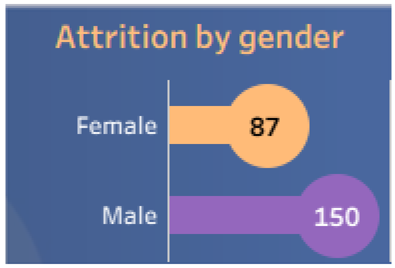



## Job Satisfaction Rating

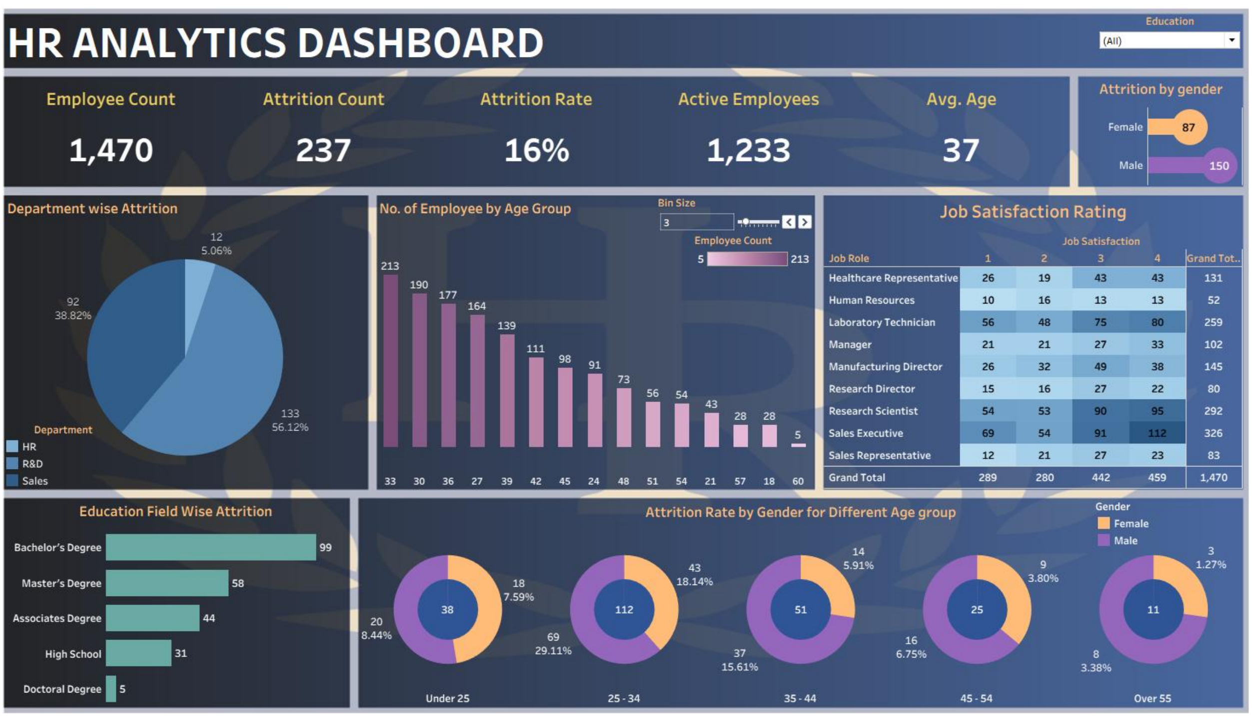
| Job Role | Job Satisfaction 1 | 2 | 3 | 4 | Grand Tot.. |
|---|---|---|---|---|---|
| Healthcare Representative | 26 | 19 | 43 | 43 | 131 |
| Human Resources | 10 | 16 | 13 | 13 | 52 |
| Laboratory Technician | 56 | 48 | 75 | 80 | 259 |
| Manager | 21 | 21 | 27 | 33 | 102 |
| Manufacturing Director | 26 | 32 | 49 | 38 | 145 |
| Research Director | 15 | 16 | 27 | 22 | 80 |
| Research Scientist | 54 | 53 | 90 | 95 | 292 |
| Sales Executive | 69 | 54 | 91 | 112 | 326 |
| Sales Representative | 12 | 21 | 27 | 23 | 83 |
| Grand Total | 289 | 280 | 442 | 459 | 1,470 |




Attrition by gender
Female
87
Male
150


Employee Count
1,470
Attrition Count
237
Attrition Rate
16%
Active Employees
1,233
Avg. Age
37


HR ANALYTICS DASHBOARD
Education
(All)
Employee Count
1,470
Attrition Count
237
Attrition Rate
16%
Active Employees
1,233
Avg. Age
37
Attrition by gender
Female 87
Male 150
Department wise Attrition
12 5.06%
92 38.82%
133 56.12%
Department
HR
R&D
Sales
No. of Employee by Age Group
Bin Size
3
Employee Count
5 213
213 190 177 164 139 111 98 91 73 56 54 43 28 28 5
33 30 36 27 39 42 45 24 48 51 54 21 57 18 60
Job Satisfaction Rating
Job Satisfaction
Job Role 1 2 3 4 Grand Tot..
Healthcare Representative 26 19 43 43 131
Human Resources 10 16 13 13 52
Laboratory Technician 56 48 75 80 259
Manager 21 21 27 33 102
Manufacturing Director 26 32 49 38 145
Research Director 15 16 27 22 80
Research Scientist 54 53 90 95 292
Sales Executive 69 54 91 112 326
Sales Representative 12 21 27 23 83
Grand Total 289 280 442 459 1,470
Education Field Wise Attrition
Bachelor's Degree 99
Master's Degree 58
Associates Degree 44
High School 31
Doctoral Degree 5
Attrition Rate by Gender for Different Age group
Gender
Female
Male
Under 25: 38; 18 7.59%; 20 8.44%
25 - 34: 112; 43 18.14%; 69 29.11%
35 - 44: 51; 14 5.91%; 37 15.61%
45 - 54: 25; 9 3.80%; 16 6.75%
Over 55: 11; 3 1.27%; 8 3.38%

## Appendix B: Sample Data (Anonymized)

The following table is a sample of 5 rows from the final, cleaned dataset used for the analysis, illustrating the data structure.

| Employee_ID | Age | Gender | Department | Job_Role | Education_Field | Job_Satisfaction | Attrition |
|---|---|---|---|---|---|---|---|
| 1001 | 41 | Female | Sales | Sales Executive | Bachelor's Degree | 2 | Yes |
| 1002 | 49 | Male | R&D | Research Scientist | Bachelor's Degree | 3 | No |
| 1003 | 37 | Male | R&D | Laboratory Technician | Master's Degree | 4 | Yes |
| 1004 | 33 | Female | R&D | Research Scientist | Master's Degree | 4 | No |
| 1005 | 27 | Male | R&D | Laboratory Technician | High School | 3 | No |

## Appendix C: Key Tableau Calculated Fields

This appendix provides the specific Tableau syntax used to generate the dashboard's primary metrics.

**[Attrition Count]**

SUM(IF [Attrition] = 'Yes' THEN 1 ELSE 0 END)

**[Active Employees]**

SUM(IF [Attrition] = 'No' THEN 1 ELSE 0 END)

OR

COUNT([Employee_ID]) - [Attrition Count]

**[Attrition Rate]**

[Attrition Count] / COUNT([Employee_ID])

(Formatted as Percentage)

**[Age Group (Categorical)]** (Used for Small Multiples)

IF [Age] <= 25 THEN 'Under 25'

ELSEIF [Age] >= 26 AND [Age] <= 34 THEN '25 - 34'

ELSEIF [Age] >= 35 AND [Age] <= 44 THEN '35 - 44'

ELSEIF [Age] >= 45 AND [Age] <= 54 THEN '45 - 54'

ELSE 'Over 55'

END

**[Dept Attrition % (Corrected)]** (Recommended fix for pie chart tooltip)

SUM([Attrition Count]) / TOTAL(SUM([Attrition Count]))

(Table Calculation computed using 'Department')